

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 June 2026
Team ID	Not Given
Project Name	Rising Waters
Maximum Marks	4 Marks

Team Members	Skill Wallet Team ID's	Roll Numbers (College: ACET)
Mohana Deepika Kancherla (Team Leader)	SWUID2026226898	23MH1A0596
Flora Disowja Sarugolu	SWUID2026212008	24P35A0549
Santhi Illa	SWUID2026212021	24P35A0550
Mathan Bhavesh Hari Raghava Krishna Munaga	SWUID2026212007	24P35A0532

Functional Requirements:

The **Rising Waters** system should securely collect weather and environmental data, including rainfall, cloud visibility, humidity, temperature, and historical flood records, to predict flood risks using machine learning models. It must provide user authentication, validate input data, and display accurate flood predictions through an interactive web dashboard. The application should store prediction history, generate reports, and integrate with external weather APIs for real-time monitoring. Additionally, it should support cloud deployment, ensure reliable performance, and deliver timely flood alerts to disaster management authorities and users.

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Register through Registration Form Register using Email Create User Profile
FR-2	User Login & Authentication	Login using Email and Password Validate User Credentials Secure Session Management
FR-3	Dashboard	Display Flood Prediction Dashboard View Prediction Summary Navigate to Prediction History
FR-4	Weather Data Input	Enter Annual Rainfall Enter Temperature, Humidity, and Cloud Visibility Enter Seasonal Rainfall
FR-5	Data Validation & Preprocessing	Validate Input Values Perform Feature Scaling using Saved Scaler Prepare Data for Prediction
FR-6	Flood Prediction	Load Trained ML Model (XGBoost) Predict Flood / No Flood Calculate Flood Probability
FR-7	Prediction Results	Display Prediction Result Show Confidence/Probability Score Provide Risk Level (Low/Medium/High)
FR-8	Prediction History	Store Prediction Records View Previous Predictions Search Prediction History
FR-9	Report Generation	Generate Prediction Reports Download Prediction History Export Results (PDF/CSV)
FR-10	User Management	Manage User Accounts Assign Roles (Admin/User) Update or Delete User Information
FR-11	Model Management	Upload Updated ML Model Manage Scaler File Monitor Model Performance
FR-12	Cloud Deployment	Deploy Application on IBM Cloud Access Application Online Ensure Secure Cloud Availability

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide a simple, intuitive, and user-friendly interface that enables users to enter weather parameters and view flood predictions with minimal training.
NFR-2	Security	The application should ensure secure user authentication, protect user data, and prevent unauthorized access through secure login and encrypted communication.
NFR-3	Reliability	The system should generate consistent and accurate flood predictions with high availability and minimal failures during operation.
NFR-4	Performance	The application should process weather inputs and display flood prediction results within a few seconds, ensuring fast response times.
NFR-5	Availability	The system should be accessible online through IBM Cloud, allowing authorized users to access flood prediction services anytime and from any location.
NFR-6	Scalability	The application should support an increasing number of users, weather datasets, and prediction requests without significant performance degradation.
NFR-7	Maintainability	The system should allow easy updates to the machine learning model, application features, and database with minimal downtime.
NFR-8	Accuracy	The system should provide highly accurate flood predictions by utilizing the best-performing machine learning model (XGBoost with approximately 96.55% accuracy).
NFR-9	Compatibility	The application should be compatible with modern web browsers and support deployment across different operating systems and cloud environments.
NFR-10	Backup & Recovery	The system should securely store prediction history and user data, with backup and recovery mechanisms to prevent data loss.